

Learning Objectives

1. Describe the important structural features of cholesterol and cholesterol esters and their physiological functions.
2. Describe the key steps of cholesterol biosynthesis and the enzymes that catalyze them.
3. Describe the mechanisms by which cholesterol biosynthesis is regulated by energy availability, hormones, food intake, and pharmacological manipulation.
4. Explain the effect of up-regulating or down-regulating plasma cholesterol levels on the intracellular biosynthesis of cholesterol and the transcriptional regulation of genes that are involved in cholesterol homeostasis.
5. Describe the biosynthesis, degradation, and regulation of bile acids; explain the clinical relevance related to bile acids metabolism.

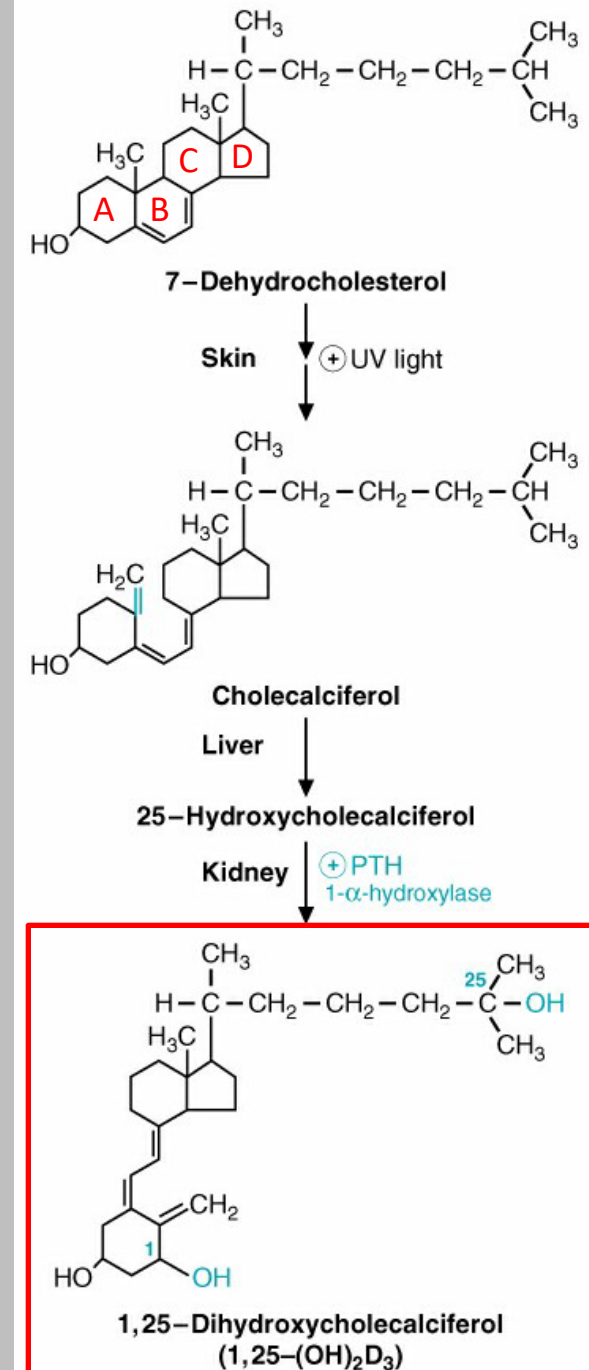
Functions of cholesterol-II:

precursor for Vitamin D biosynthesis

Vitamin D₃ and calcium metabolism:

- Vitamin D₃ is converted to a hormone that enhances intestinal Ca²⁺ absorption and demineralization of the bone
- Vitamin D₃ is synthesized endogenously from cholesterol in skin when exposed to UV
- 1,25-Dihydroxycholecalciferol is the active form of Vitamin D₃

PTH: parathyroid hormone



Functions of cholesterol-III: precursor for steroid hormone biosynthesis

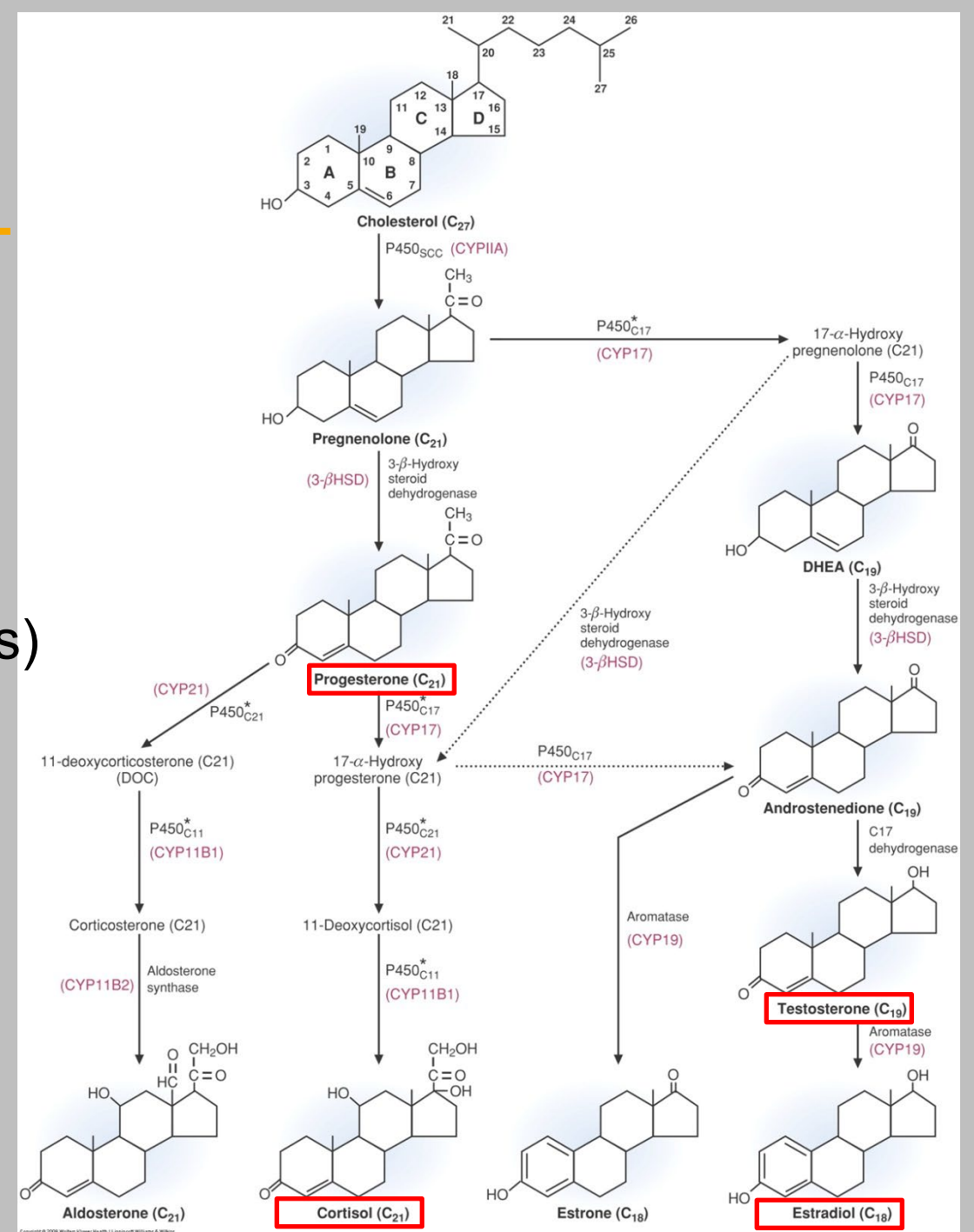
Synthesis:

- Shortening the hydrocarbon chain
- Oxidation of the steroid nucleus

Steroid hormones:

(sex steroids, glucocorticoids, mineralocorticoids)

- Progesterone
- Testosterone
- Estradiol
- Cortisol
- Others

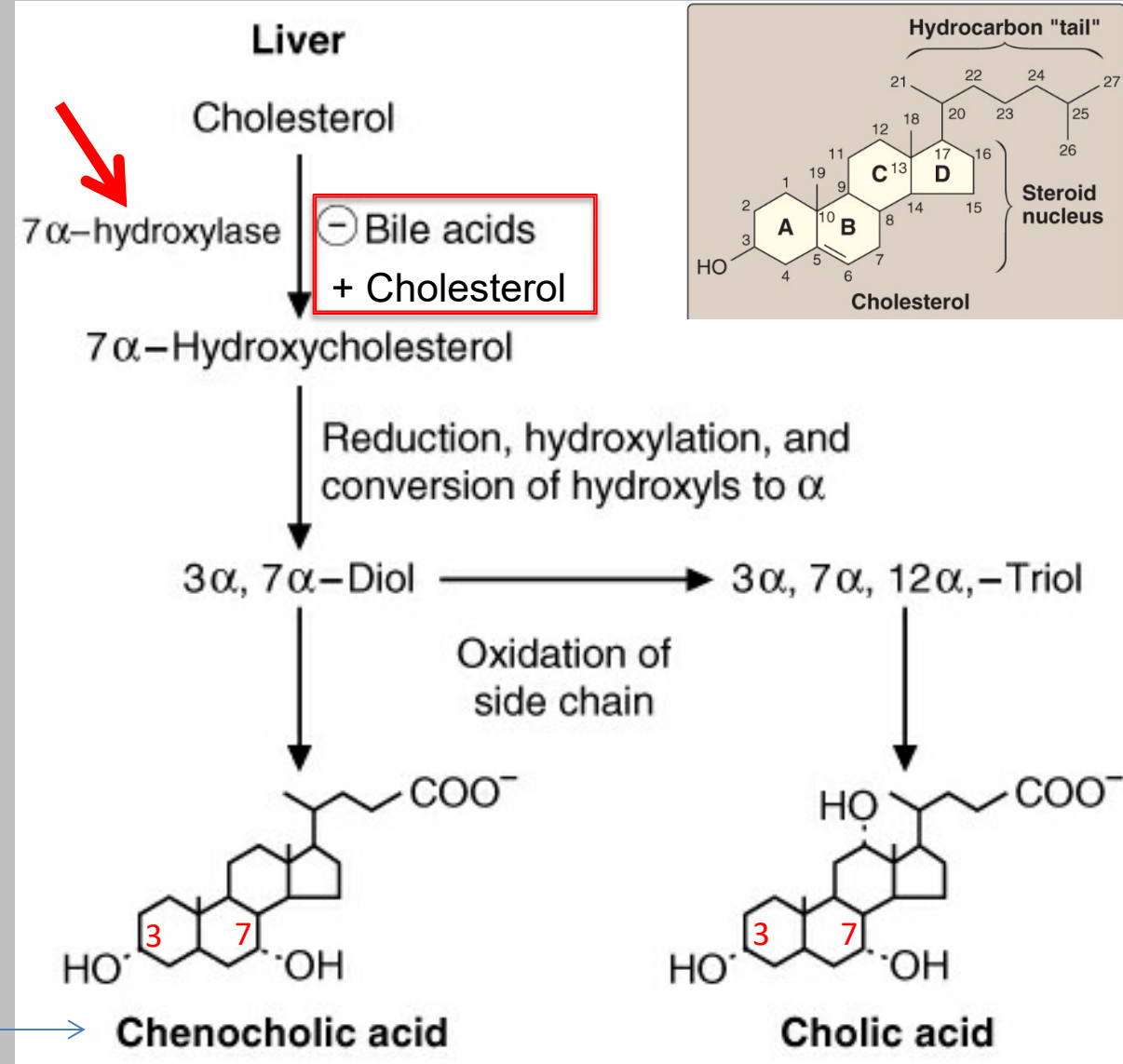


Functions of cholesterol-IV:

precursor for bile acids/salts biosynthesis

Quantitatively, the most important product of cholesterol are **bile acids**.

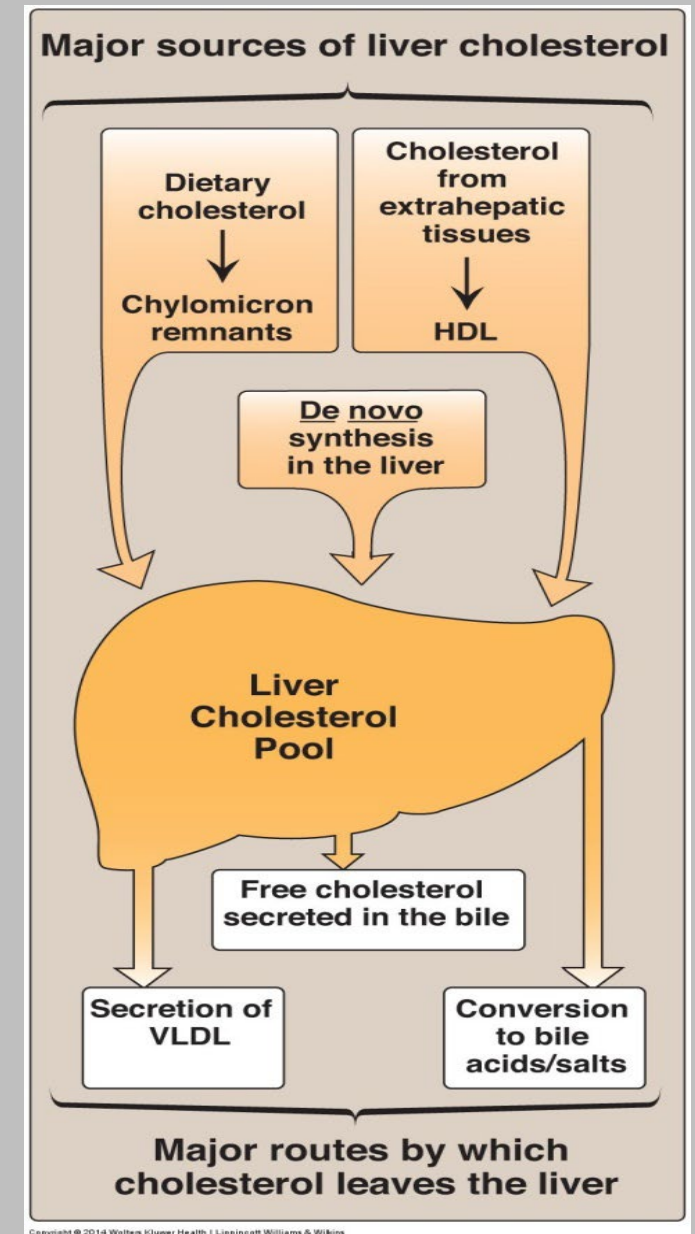
7-hydroxylase catalyzes the **rate-limiting step** of bile acid synthesis.



Primary bile acids:

Cholesterol Balance

- A typical Western diet contains roughly **400-600 mg /day** of cholesterol; about half is absorbed, which is mediated by the Niemann-Pick C1-like 1 protein (**NPC1L1**), the target of **ezetimibe**.
- About **800 mg / day** of cholesterol is synthesized by the body.
- This production is balanced mostly by the excretion of **bile acids**, about **1000-1100 mg/ day**.
- So, if we want to decrease cholesterol levels in the body, we could:
 1. Decrease **dietary** cholesterol uptake
 2. Decrease **cholesterol biosynthesis**
 3. Increase the **excretion** of bile acids





The structure of my lectures related to key metabolic pathways

- 1) What is the **purpose** or **importance** of the pathway?
- 2) What is (are) the **committed**, or **rate-limiting**, or **irreversible** reaction(s)? The **enzyme**(s)?
- 3) What are the **featured** reactions of the pathway and enzymes catalyzing the reactions?
- 4) How is the pathway **regulated**?
- 5) **Clinical correlation.**

Biosynthesis of cholesterol

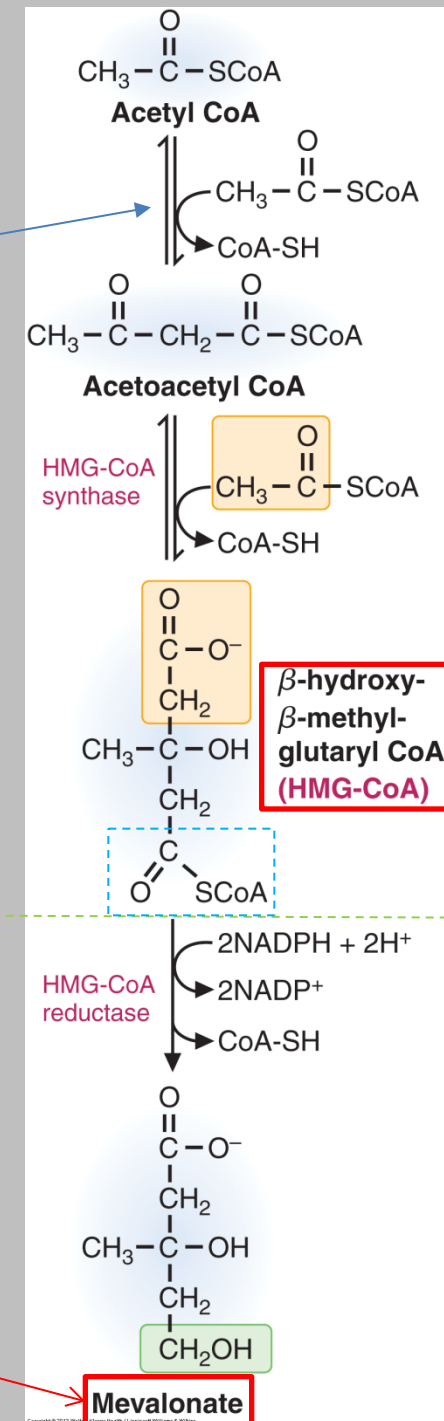
- Most tissues can synthesize cholesterol, including **liver** (the **main source**), adrenals, gonads, intestine, placenta.
- Biosynthesis is metabolically regulated.

Biosynthesis of cholesterol: Stage 1 (continued)

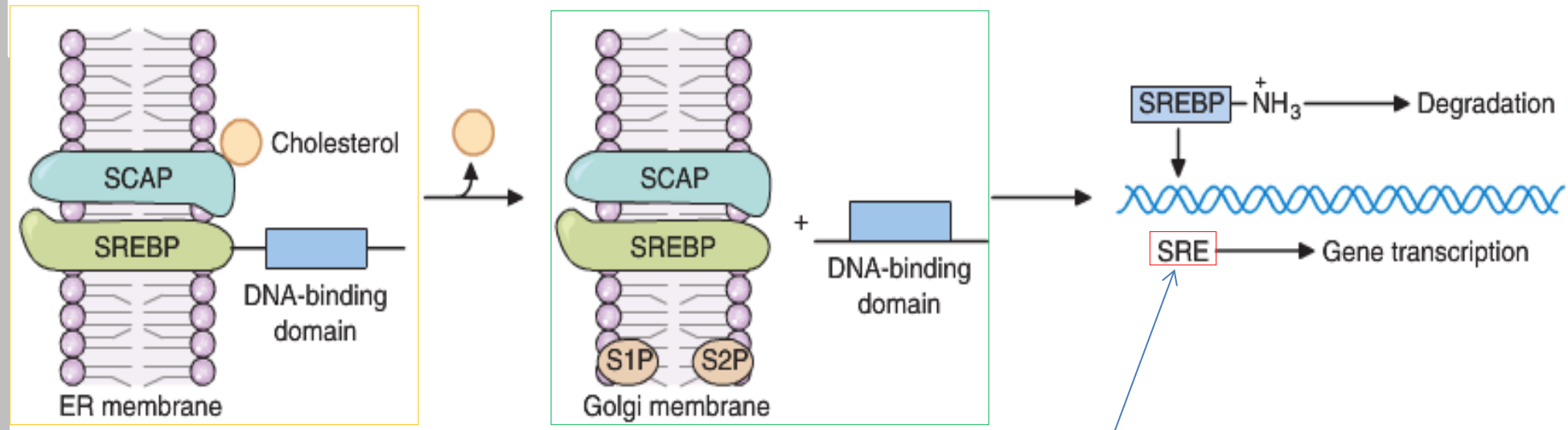
Stage 1: Synthesis of **mevalonate (6C)** from **acetyl CoA (2C)**

1. Synthesis of **acetoacetyl CoA (4C)** from two molecules of **acetyl CoA** by **thiolase**
2. Cytosolic **HMG CoA synthase** combines **acetoacetyl CoA** with another molecule of **acetyl CoA**
3. **HMG CoA reductase** catalyzes the **rate-limiting step** and is the **key regulatory** point in cholesterol biosynthesis.
 - Reduces HMG CoA and produce one **mevalonate**
 - Requires **2X NADPH**
 - Located on the **ER** membrane
 - **Irreversible**

3X Acetyl CoA



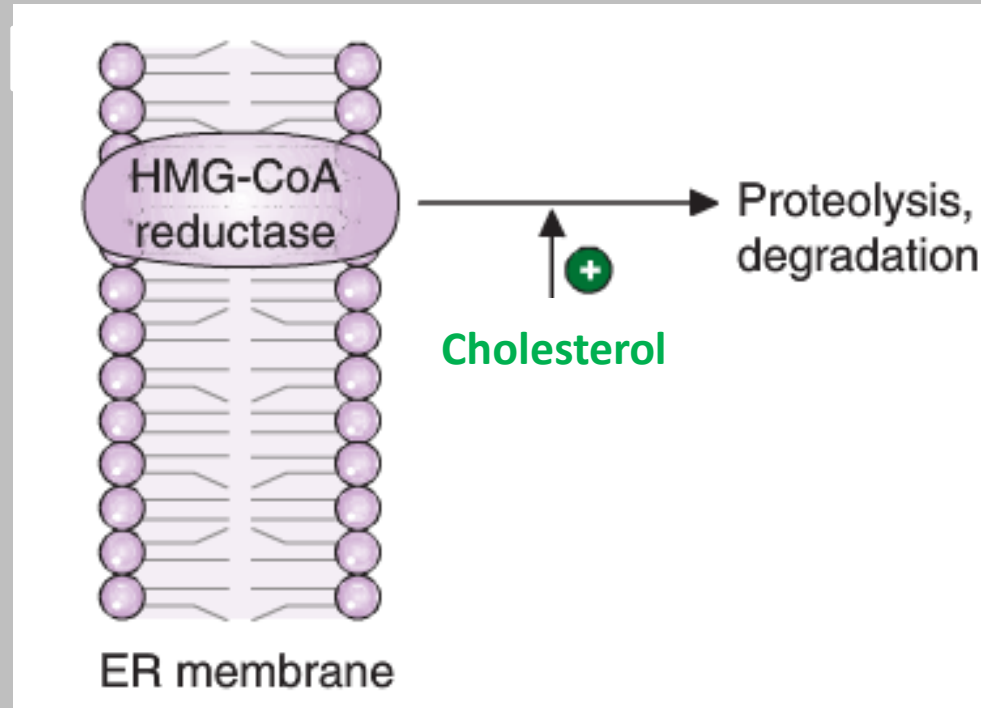
Regulation of cholesterol biosynthesis-I: via transcription



Transcriptional regulation of cholesterol biosynthesis:

- Sterol Regulatory Element Binding Protein (**SREBP**) regulates the transcription of **HMG CoA reductase**
- SREBP DNA-binding domain binds to the Steroid Response Element (**SRE**) of the HMG CoA reductase gene and turns on its transcription when cholesterol levels are low
- **SREBP** is an ER membrane protein and binds **SCAP** (SREBP Cleavage Activating Protein)
 - when [**Cholesterol**] is **high**, cholesterol binds to **SCAP**, which then traps SREBP at the ER.
 - when [**Cholesterol**] is **low**, **SCAP-SREBP** is transported to **Golgi**
- In **Golgi**, **SCAP** activates certain **proteases**, which then cleave the **SREBP** and release its DNA binding domain, which then translocates to the nucleus, binds **SRE**, and **activates** the transcription of genes involved in cholesterol biosynthesis, including HMG CoA reductase.

Regulation of cholesterol biosynthesis-II: via degradation



Regulation of cholesterol biosynthesis at the HMG CoA reductase step:

Post-transcriptional control when cholesterol is abundant:

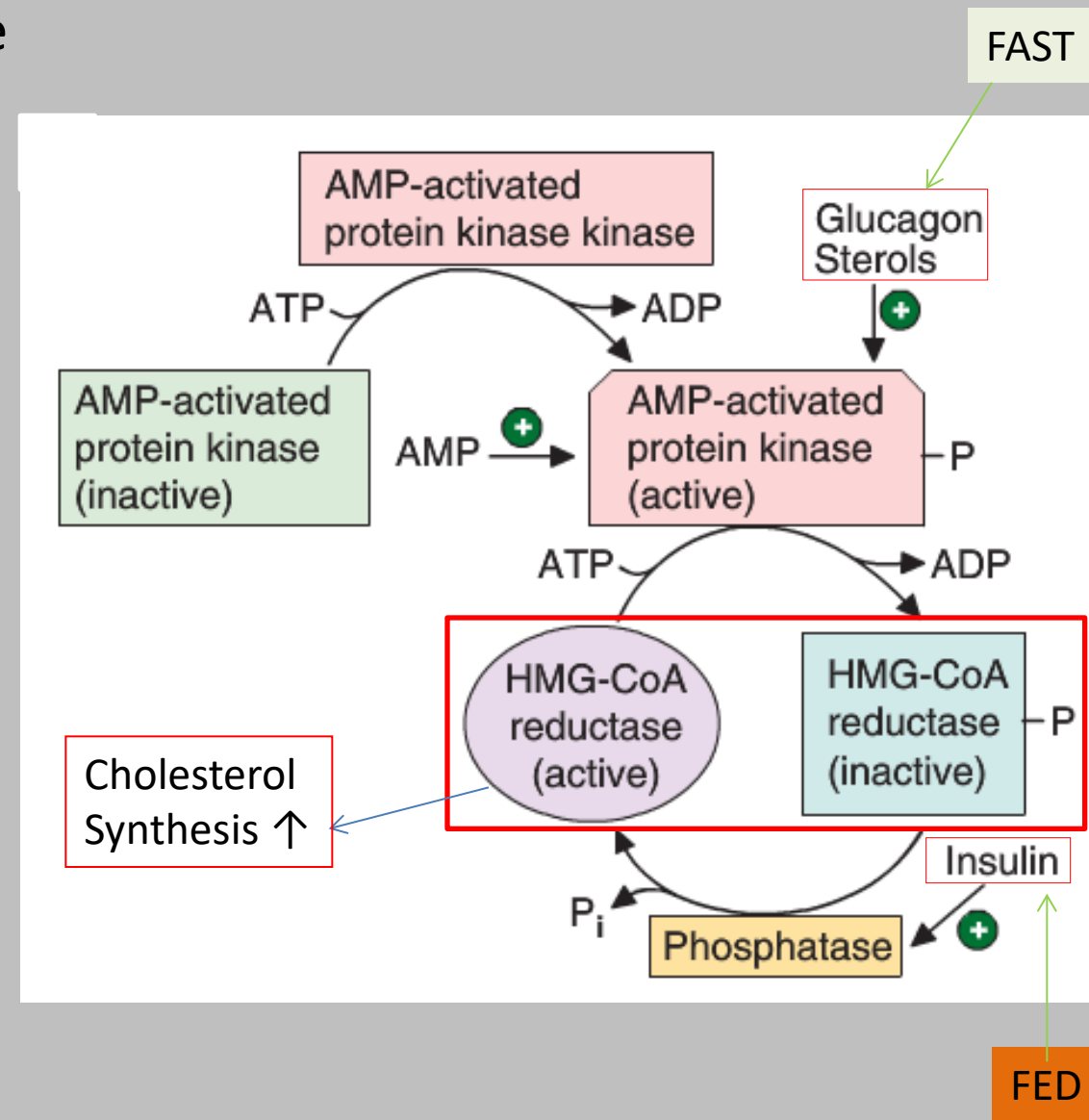
- HMG CoA reductase **mRNA** is targeted for degradation
- HMG CoA reductase **protein** is targeted for degradation by the proteasome

Regulation of cholesterol biosynthesis-III:

phosphorylation/dephosphorylation

Regulation of cholesterol biosynthesis through the phosphorylation and dephosphorylation of HMG CoA reductase:

- AMP ↑ (low energy signal) → AMP-activated protein kinase (**AMPK**) ↑ → phosphorylation ↑
- **Glucagon** and high level of sterols can activate AMPK → phosphorylation ↑
- Phosphorylation of HMG CoA reductase reduces its activity.
- **Insulin** activates a protein phosphatase, resulting in the dephosphorylation of HMG CoA reductase, which increases its activity.



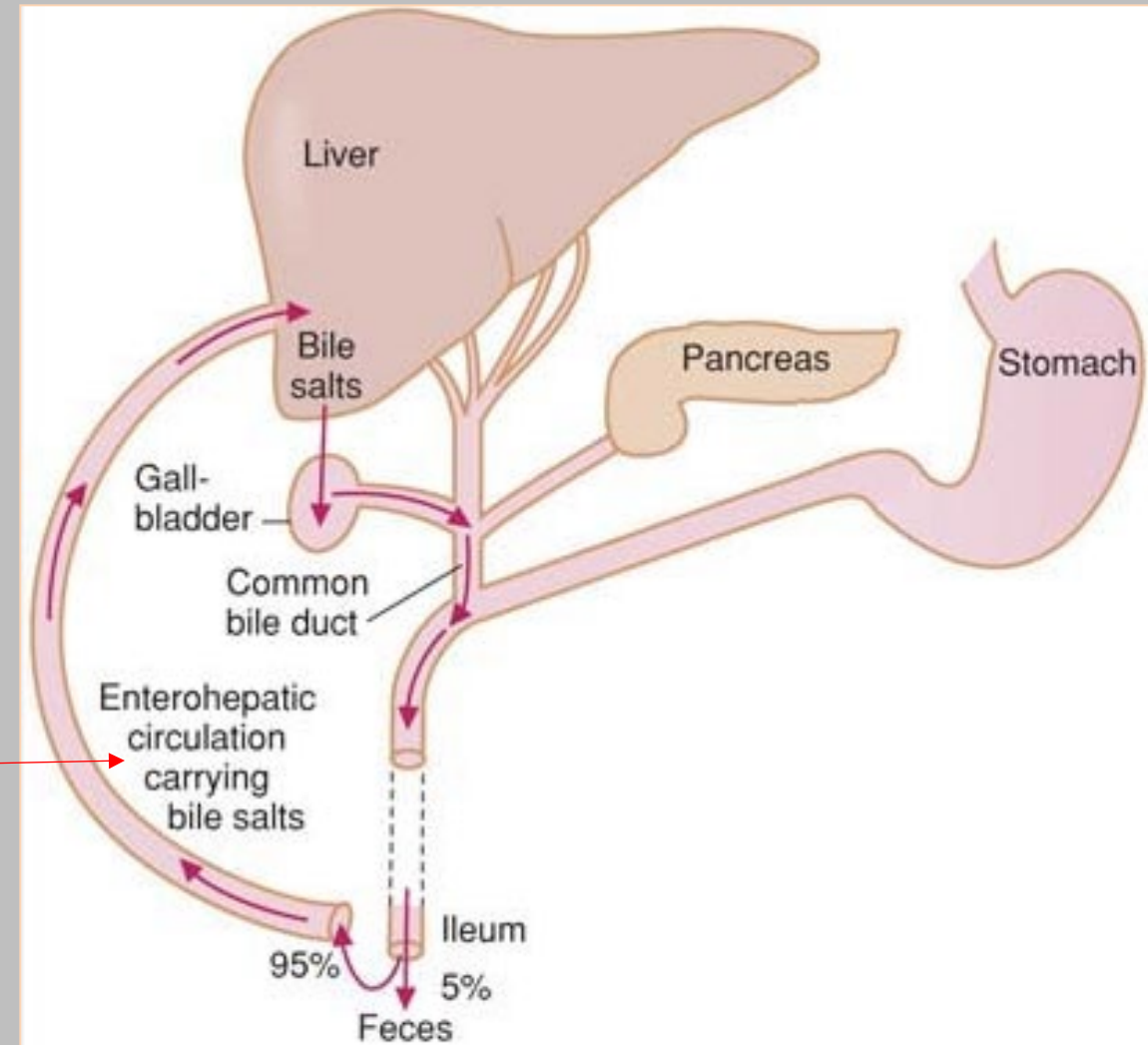


Summary of the regulation of cholesterol biosynthesis - HMG CoA reductase

- Catalyzes the **rate-limiting step**
- **Cholesterol** functions as a **feedback inhibitor** of HMG CoA reductase synthesis through a delicate **transcriptional** regulation system: **SREBP-SCAP** is retained in the ER membranes when cholesterol level is high. When cholesterol is low, **SREBP-SCAP** is transported to Golgi; at Golgi, **SREBP** is cleaved and the cleavage fragment of **SREBP** enters the nucleus to serve as a transcription factor and stimulates the expression of HMG CoA reductase and other genes
- Both HMG CoA reductase mRNA and protein can be **degraded** by when cholesterol level is high.
- **AMP**, a low energy indicator, controls HMG CoA reductase activity through phosphorylation (by **AMPK, inactive**) and dephosphorylation (**active**).
- **Insulin** and **thyroid hormone** **stimulate** HMG CoA reductase activity; **glucagon, cholesterol, chylomicron remnants**, and **Low-density lipoproteins (LDL)** **inhibit** HMG CoA reductase activity
- HMG CoA reductase is the molecular target of cholesterol lowering drugs - **statins**

The role of bile salts in the digestion and absorption of lipids

- Dietary lipids (fat) have low solubility
- Bile salts act as the **detergent** to emulsify dietary lipids
- Bile is a mixture of water, phosphatidylcholine (**lecithin**), bile salts, and cholesterol
- Cholesterol is the **precursor** of bile salts
- Bile salts are synthesized in the **liver** and stored in the gallbladder
- Bile salts can be reabsorbed in the GI tract for re-circulation back to the liver (**enterohepatic circulation**)
- A significant amount of bile salts are excreted in the feces everyday. Bile salts excretion is an important way to reduce **cholesterol** level



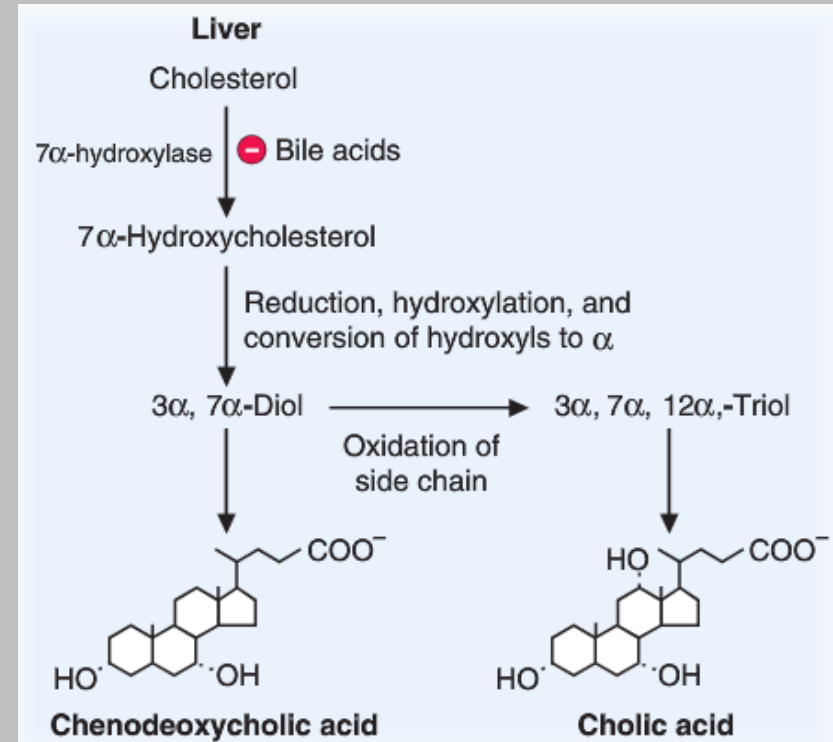
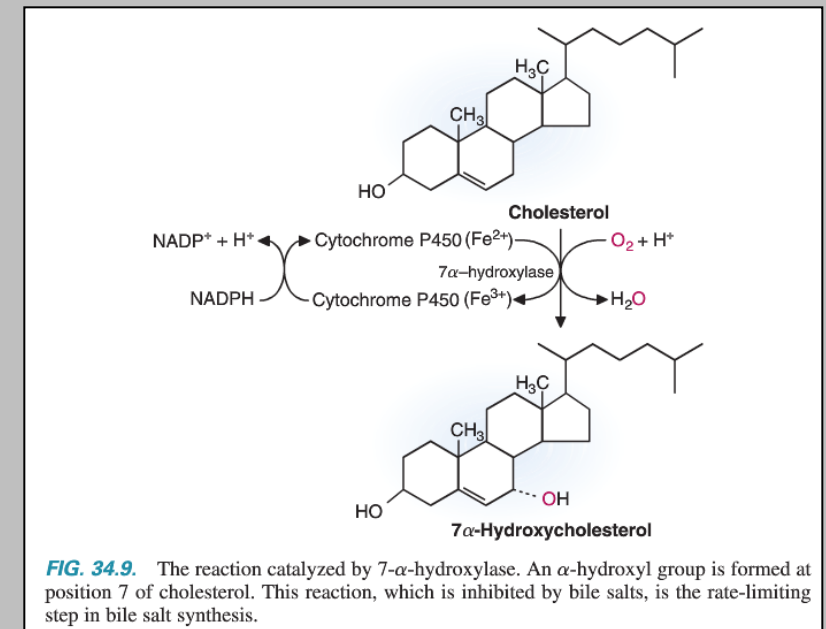
Synthesis of bile salts-I

Bile acids (**24C**) are **synthesized** from cholesterol (**27C**) by:

- increasing **hydroxylation** of the steroid nucleus
- removing **three carbons (-3C)** from the aliphatic chain
- adding a **carboxyl group** at carbon 24
- **reducing** the double bond in cholesterol

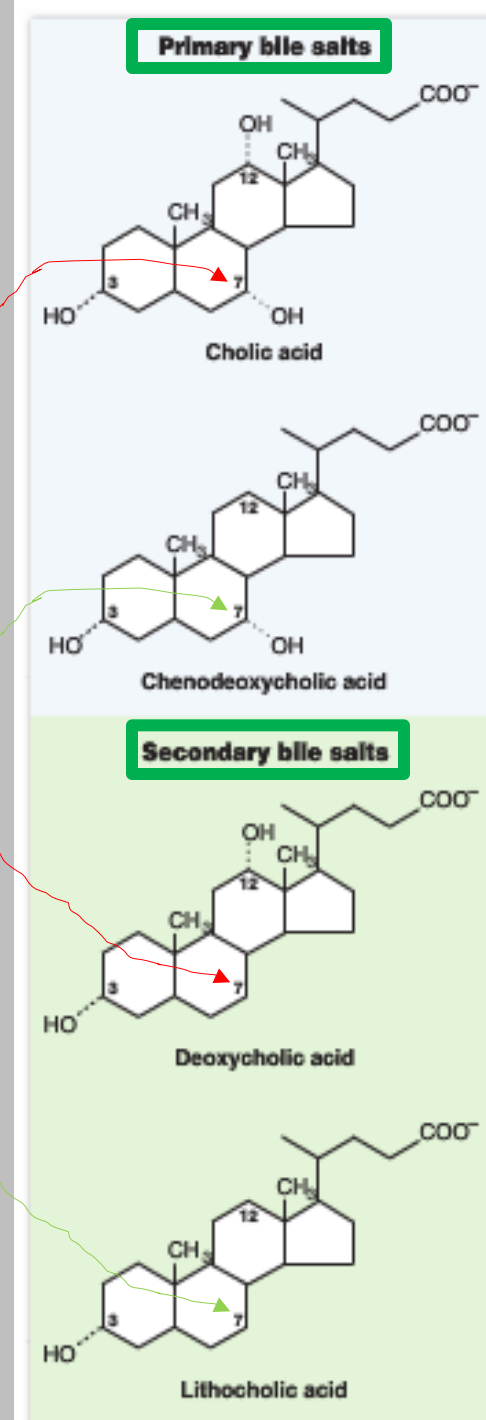
Hydroxylation and **carboxylation** increase the solubility of bile salts.

Primary bile acids:

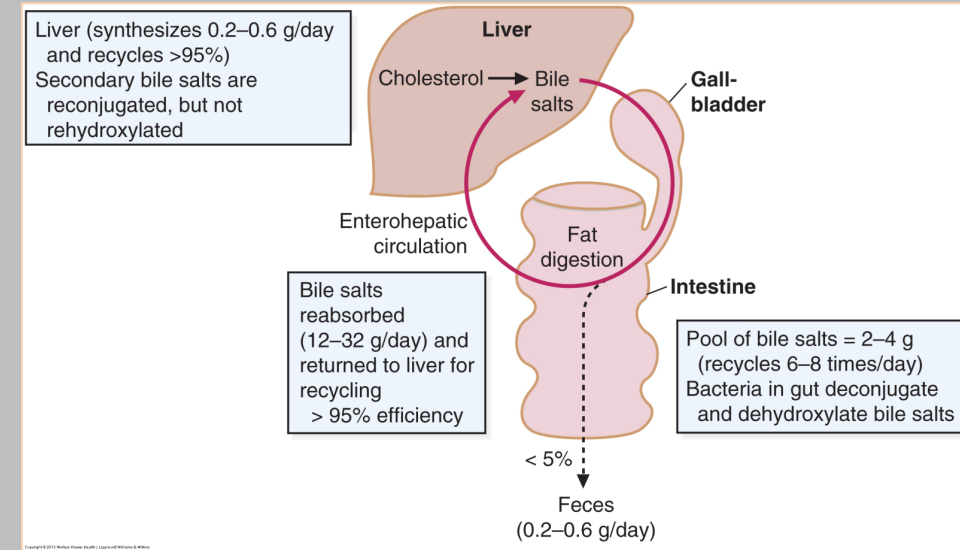


Synthesis of bile salts-II

- Bile acids are synthesized in the **liver** directly from cholesterol; **cholic acid** and **chenodeoxycholic acid** are also known as the 'primary bile acids'
- Hydroxylation** at carbon 7 by **7 α -hydroxylase** is the **rate-limiting step**
- Deoxycholic acid** and **lithocholic acid** are formed in the intestine from the primary bile acids by intestinal **bacterial enzymes** and are thus called the 'secondary bile acids'
- Secondary bile salts** are **less soluble** (because they lost the OH groups at carbon-7) and more likely to be excreted



Summary of bile salts metabolism



- The bile acids secreted into the intestine are efficiently re-absorbed back into the blood stream at the ileum (**> 95%**) and subsequently return to the liver.
The liver → bile duct → duodenum → ileum → portal vein → liver cycle of bile acids is called the **enterohepatic circulation**.
- The efficiency of the liver in extracting bile acids may be compromised under conditions of liver disease, resulting in increased bile acids in the blood.
- decreasing the re-absorption of bile salts can be a viable strategy to reduce the level of cholesterol, for example, by bile salts sequestrant (**cholestyramine**) and dietary fibers.

A note: In medical terms, "entero" is a prefix that means "intestine". It comes from the Greek word *énteron*, which also means "intestine".

Summary

- **Key concepts:** steroid, sterol, steroid nucleus, plant sterols, bile acids/bile salts, primary bile salts, secondary bile salts, conjugated bile salts, enterohepatic circulation
- Physiological function of cholesterol and bile acids
- The rate-limiting step of cholesterol and bile acids biosynthesis and the key enzymes involved
- Key regulatory mechanisms of cholesterol and bile acids biosynthesis
- **Clinical correlations** (disease symptoms, biochemical basis, and treatment): various strategies to lower cholesterol level; cholelithiasis